THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · EASY LEVEL UNIT TEST SET 1

Chapters 1–4: **Chemical Reactions and Equations, Acids, Bases and Salts, Metals and Non-metals, Carbon and its Compounds** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A chemical change always produces: 1

(a) only a change of colour **(b)** only a change of state
(c) no new substance **(d)** one or more new substances

2. The taste of acids is generally: 1

(a) sweet **(b)** sour **(c)** bitter **(d)** salty

3. The property of metals by which they can be beaten into thin sheets is: 1

(a) lustre **(b)** malleability **(c)** sonority **(d)** ductility

4. The atomic number of carbon is: 1

(a) 4 **(b)** 8 **(c)** 12 **(d)** 6

5. $2\,H_2 + O_2 \longrightarrow 2\,H_2O$ is a: 1

(a) combination reaction **(b)** double displacement reaction
(c) decomposition reaction **(d)** displacement reaction

6. The pH of a neutral solution is: 1

(a) 7 **(b)** 14 **(c)** 1 **(d)** 0

7. The metal least reactive among the following is: 1

(a) zinc **(b)** gold **(c)** iron **(d)** sodium

8. A single bond between two carbon atoms is found in: 1

(a) ethane **(b)** ethene **(c)** ethyne **(d)** benzene

9. The colour of copper sulphate solution is: 1

(a) blue **(b)** yellow **(c)** green **(d)** colourless

10. Acid + base gives: 1

(a) only water **(b)** salt and water
(c) only salt **(d)** salt and hydrogen

11. The ore of iron is: 1

(a) galena **(b)** haematite **(c)** cinnabar **(d)** bauxite

12. Diamond and graphite are: 1

(a) isomers **(b)** allotropes of carbon
(c) alloys **(d)** compounds

13. Substances on the left side of a chemical equation are called: 1

(a) products **(b)** precipitates **(c)** catalysts **(d)** reactants

14. Bases generally feel: 1

(a) rough **(b)** soapy **(c)** sticky **(d)** oily

15. The property of metals by which they can be drawn into wires is: 1

(a) ductility **(b)** malleability **(c)** conductivity **(d)** sonority

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. State the law of conservation of mass. 2

17. Define an acid and give two examples. 2

18. Define malleability and ductility. 2

19. What is the valency of carbon? Why? 2

20. Define a decomposition reaction and give an example. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Balance the equation $\mathrm{Na} + \mathrm{Cl_2} \longrightarrow \mathrm{NaCl}$. 3

22. Write the colours of red litmus, blue litmus and phenolphthalein in 3

(a) lemon juice

(b) soap solution.

23. Write any three physical properties of metals. 3

24. Write the molecular formulas of the first three alkanes with names. 3

OR

Name the type of reaction: $\mathrm{CaO} + \mathrm{H_2O} \longrightarrow \mathrm{Ca(OH)_2}$. Give a reason and say whether it is exothermic or endothermic. 3

25. State three properties of acids. 3

OR

Write the equation for sodium with water and name the products. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. A magnesium ribbon is burnt in air. State the observation, write the balanced equation, name the type of reaction and the product. Why is the ribbon cleaned first? 5

OR

State five differences between acids and bases (taste, litmus, ions, pH, example). 5

27. Write five differences between metals and non-metals. 5

OR

Draw electron dot structures of methane, ethene and water. State number of bonds. 5

END OF QUESTION PAPER — **Total: 50 Marks**

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